

Identification of Surface Basic Sites and Acid–Base Pairs of Hydroxyapatite

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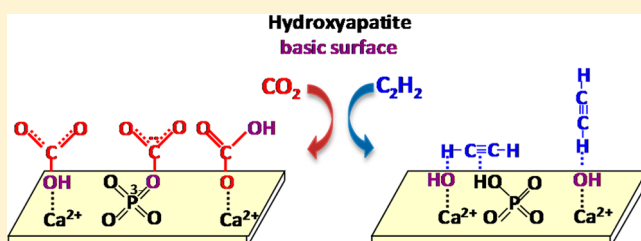
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Supporting Information

ABSTRACT: The Lewis and Brønsted basic properties of a stoichiometric hydroxyapatite (HAp) were investigated by infrared spectroscopy following the adsorption and desorption processes of a Lewis acidic molecule, CO₂, and a Brønsted acidic molecule, C₂H₂. CO₂ interacts with basic OH[−] and O^{2−} of PO₄^{3−} groups to form hydrogenocarbonates and surface carbonates, respectively. It also generates surface type A carbonates and related water upon substitution of two neighboring structural OH[−] groups. Water modifies the basic properties of the HAp by decreasing the surface carbonatation and enhancing the formation of hydrogenocarbonates, and promotes the substitution ability of OH[−] by carbonates. Due to the affinity of HAp for carbonatation, the thermodesorption experiment of CO₂ accounts for the thermal decomposition of bulk type A and B carbonates rather than for the lone surface basicity. As for the acetylene probe, three nondissociative adsorption modes of acetylene on the HAp surface are observed: a π complex interaction with acidic POH, an interaction with an acid–base (POH–OH) pair, and finally, a σ complex interaction with basic OH[−] that is the most stable upon desorption. There is no evidence of the involvement of basic O^{2−} of PO₄^{3−} in the interaction with acetylene. It is thus proposed that both acidic POH and basic OH[−] groups may play a determinant role in acid–base properties of hydroxyapatites.



1. INTRODUCTION

In the context of increasing demand in fossil energy and of the depletion of petroleum resources,^{1,2} there is a growing interest in the use of renewable energy sources, such as biomass. Such an energetic transition points toward new emerging challenges for the valorization of new classes of molecules. Such an evolution together with the increasing environmental constraints also concerning fine chemistry resulted in a surge of interest in solid base-catalyzed reactions. Indeed, besides the well-known efficiency and selectivity of solid-base catalysts for isomerization, Knoevenagel condensation, and Michael reactions, these basic materials can also be of particular interest for energy facilities, especially for the valorization of alcohols from biomass.^{3–8} Recent studies dealing with alkaline earth oxides^{9,10} or hydrotalcites^{11–13} pointed out the peculiar role of weak basic OH[−] groups in basic reactivity.^{9,11–13} This led us to investigate the hydroxyapatite (HAp) system (Ca₂(PO₄)₆(OH)₂), which includes both PO₄^{3−} and OH[−] groups possibly acting as weak basic sites. This system is quite complex due to its large versatility of composition, which is related to its peculiar ability toward structural substitutions on calcium sites but also on OH[−] or PO₄^{3−} sites (by CO₃^{2−}, for instance, respectively, referred to as type A and B carbonates).¹⁴ The HAp system is very promising for acid–base reactions since it has been shown

to be efficient for many base-catalyzed reactions^{15–18} and very selective in the formation of *n*-butanol from ethanol (Guerbet reaction).^{8,19–22} Yet, little is known about how HAp proceeds at a molecular level. The stoichiometry, a macroscopic parameter, that is related to the calcium relative content,¹⁴ has been shown to influence the acid–base properties of hydroxyapatites to a large extent. The lowering of the Ca/P ratio resulted in a decrease in the basic yield.²³ Even if the calcium deficiency (Ca_{2−*x*}(HPO₄)_{*x*}(PO₄)_{6−*x*}(OH)_{2−*x*}) is associated with a decrease in the number of both PO₄^{3−} and OH[−] groups and with the appearance of HPO₄^{2−} groups to achieve the charge balance, no direct evidence of the involvement of surface basic OH[−] groups of hydroxyapatites in the catalytic process of base reactions has been provided to date.

Adsorption of probe molecules is very commonly used to investigate the surface acid–base properties of solids at the gas–solid interface.^{24–30} Infrared allows to reveal what are the nature and structure of the adsorbates formed.^{24,25,31} In some cases, the nature of the adsorption site can also be identified.³² CO₂ is the most common probe used to investigate the basic

Received: January 15, 2014

Revised: May 9, 2014

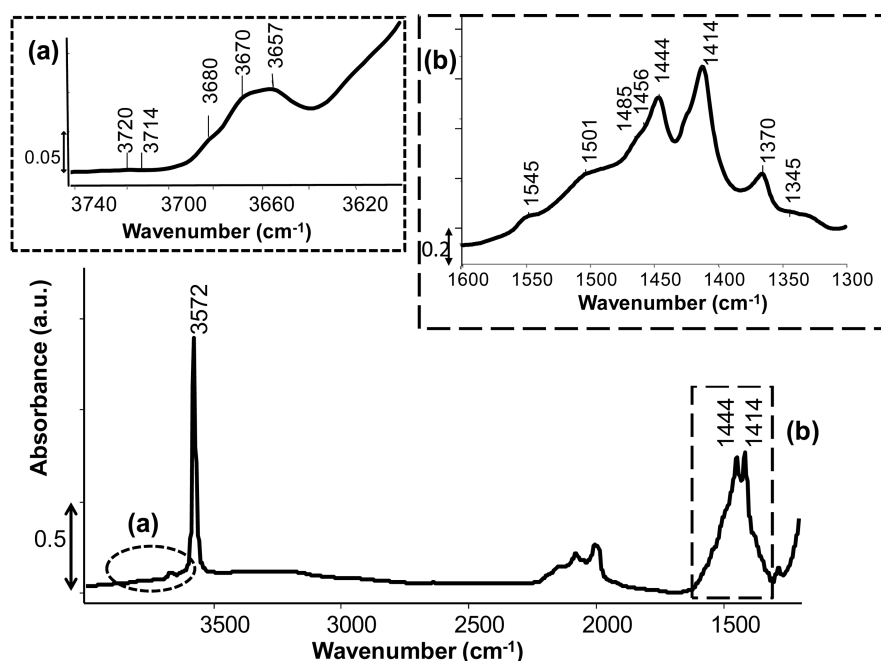


Figure 1. FTIR spectrum of a self-supported wafer of the stoichiometric HAp recorded at RT after in situ pretreatment under vacuum at 623 K: (a) Zoom in the 3740–3600 cm^{-1} area, (b) zoom in the 1650–1300 cm^{-1} area.

properties at the gas–solid interface.^{4,24,29,33} It has been used earlier to characterize the basicity of hydroxyapatites, but mainly via thermo-desorption or calorimetric measurements.^{34–36} To our knowledge, the very few infrared data dealing with the interaction of CO_2 with hydroxyapatites^{35,37} did not report on the basic properties of hydroxyapatites.

Complementary to this Lewis basicity probe molecule, protic molecules have also been proposed to characterize the Brønsted basicity (ability to attract a proton).^{38–44} Even though these molecules exhibit several adsorption modes, dissociative and/or molecular, and can probe both acidic or basic sites, they can also adsorb on an acid–base pair.^{38–42} This may help to reveal the peculiar acid–base pairs required for many basic catalytic reactions. Indeed, the first step of most basic reactions consists in a deprotonation and the formation of an anionic intermediate adsorbed on the acidic site close to the basic entity.⁴⁰ Acetylenic compounds have been shown to be sensitive to the strength of the basic sites^{32,45–47} but have never been used to investigate a HAp system to date.

The aim of the present paper is to investigate by infrared spectroscopy how Lewis acidic CO_2 and Brønsted acidic acetylene molecules can be accounted for the characterization of the Lewis and Brønsted basicities respectively, of a stoichiometric hydroxyapatite surface. The adsorption and desorption processes of CO_2 on a hydroxyapatite are followed by FTIR. The isotopic labeling of the surface by D_2O prior to CO_2 adsorption allows a better description of the CO_2 adsorption processes and to discuss the influence of chemisorbed water on the basicity. Acetylene adsorption on a hydroxyapatite is studied by DRIFTS. The nature of the surface sites interacting with these two probe molecules is discussed, leading us to propose an acid–base pair possibly involved as an active site in basic catalytic reactions catalyzed by hydroxyapatites.

2. EXPERIMENTAL SECTION

2.1. HAp Preparation. A stoichiometric calcium hydroxyapatite sample ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) was prepared according to the procedure previously described:²³ the coprecipitation of solutions of $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{HPO}_4$ previously adjusted at pH = 10 was performed at 353 K under a N_2 flow to limit carbonation of the material. The pH was maintained at pH = 10 during the precipitation and maturation time by periodic addition of NH_3 . The washed precipitate was then dried overnight at 373 K and thermally treated under an Ar flow (150 mL min^{-1}) up to 623 K (5 K min^{-1}) and maintained at this temperature for 90 min.

From elemental analysis performed by ICP-AES by the “Service Central d’Analyse” of the CNRS (Solaize, France), the Ca and P contents were determined and confirmed that the sample was stoichiometric (Ca/P ratio of ~ 1.67). The C content was found to be about 0.5 wt %. It was also checked from XRD (Siemens diffractometer equipped with a Copper anode generator ($\lambda = 1.5418 \text{ \AA}$)) that the sample exhibited the hydroxyapatite structure (ICDD pattern 01–074–9780(A)). Its specific surface area, measured by N_2 adsorption at 77 K on a Micromeritics (ASAP 2010) apparatus, was estimated to be 39 $\text{m}^2 \text{g}^{-1}$ from the BET method.

2.2. FTIR. The sample was pressed at 2.5 $\text{ton}\cdot\text{cm}^{-2}$ into self-supporting wafers (20–25 mg, 16 mm diameter). Wafers were placed in a quartz cell equipped with ZnSe windows and connected to a vacuum line allowing thermal treatments and adsorption–desorption experiments to be carried out in situ. The FTIR spectra were recorded using a Brüker FTIR Vertex 70 spectrometer, equipped with a MCT detector (2 cm^{-1} resolution, 64 scans per spectrum).

The wafers were pretreated under an Ar flow (20 mL min^{-1}) up to 623 K (5 K min^{-1}) for 90 min before being evacuated at 623 K (10^{-4} Pa). The interaction of CO_2 with the surface was monitored at room temperature (RT) while adding incremental doses of CO_2 up to a final equilibrium pressure of 133 Pa. An evacuation step was then carried out starting from RT up to

523 K until a residual pressure of about 10^{-4} Pa was reached. Additional set of data were obtained from CO_2 adsorption on a HAp surface that had been previously partially deuterated following the surface isotopic labeling procedure already reported elsewhere: D_2O (266 Pa) was adsorbed at 373 K for 10 min on the pretreated sample before the sample was outgassed at this temperature for 10 min. This cycle was repeated twice. An additional evacuation step at 473 K was then performed for 20 min to remove most of the residual chemisorbed D_2O .⁴⁸ The related partially deuterated surface will be hereafter referred to as deuterated sample.

Unless specified otherwise, the FTIR spectra of adsorbed molecules are reported in absorbance, after subtraction of the background of the sample before the adsorption. The two series of spectra recorded on the nondeuterated and deuterated samples have been normalized using the overtone and combination modes of the P–O bands ($2200\text{--}1900\text{ cm}^{-1}$).

2.3. DRIFTS. Diffuse reflectance infrared spectra were recorded using a Brüker IFS 66 V spectrometer (4 cm^{-1} resolution, 256 scans/spectrum, MCT detector). About 40 mg of powdered sample, corresponding either to pure HAp or to a mechanical mixture of 10 mg of HAp in 90 mg of diamond powder, was placed inside a heated crucible located in a Thermo Spectra-Tech high temperature cell equipped with two ZnSe windows and with appropriate gas inlet and outlet connections as to pass the gas flow through the catalytic bed.

To investigate the interaction of HAp with acetylene, the sample was heated to 623 K (5 K min^{-1}) under Ar (20 mL min^{-1}) for 90 min and then cooled down under Ar to RT. The gas flow was switched to 5% acetylene in Ar (20 mL min^{-1}) for 10 min to follow the acetylene adsorption and then switched back to pure Ar to follow acetylene desorption at RT. The DRIFT data recorded upon acetylene adsorption also involve a gas phase contribution of acetylene that was systematically subtracted from the recorded spectra.

The reference spectrum is recorded with KBr (Fluka, purity > 99.5%). The DRIFT spectra of acetylene adsorption are reported in $\text{Log}(1/R')$ with the relative reflectance $R' = I_{\text{HAp+acetylene}}/I_{\text{HAp}}$ (spectrum recorded after adsorption of acetylene/spectrum recorded just before acetylene adsorption).⁴⁹

3. RESULTS AND DISCUSSION

3.1. Infrared Spectra of the Activated Samples. The typical FTIR spectrum of the stoichiometric HAp recorded after pretreatment at 673 K is presented in Figure 1. A similar spectrum is obtained from DRIFTS (see Supporting Information, Figure S1). In the $\nu_{\text{(OH)}}$ region, the main band is observed at 3572 cm^{-1} . It was assigned to both the contributions of structural OH^- groups located inside the channels (bulk OH^-) and to structural OH^- emerging at the surface (surface OH^-).^{48,50} Bands of weaker intensity (inset (a) in Figure 1) are detected at higher wave numbers. They are related to PO–H vibrators^{35,51,52} associated either with bulk defective HPO_4^{2-} species (band at 3657 cm^{-1})⁴⁸ or with surface protonated terminal phosphate groups (bands at 3720, 3714, 3680, 3670 cm^{-1})⁴⁸ that achieve the surface charge balance.^{53,54}

Besides a set of weak components at the $2200\text{--}1900\text{ cm}^{-1}$ related to the overtone and combination modes of the P–O bands (the corresponding stretching mode below 1350 cm^{-1} cannot be observed due to their high absorbance), the broad multicomponent contribution observed at $1550\text{--}1300\text{ cm}^{-1}$

(inset (b), Figure 1) is assignable to the presence of carbonate groups (Figure 1).^{55,56} Their assignment is still controversial, probably due to the fact that the conditions of collection of the spectra greatly impact the shape of the spectra, as illustrated in Supporting Information (Figure S2) for FTIR spectra of a same HAp sample recorded either from dilution in KBr or from thermally treated self-supported pellet. According to Cheng et al.³⁷ who studied self-supported wafers, the most intense contributions at 1444 and 1414 cm^{-1} , as well as the weak contribution at 1501 cm^{-1} , are assigned to the incorporation of bulk carbonates located in the A sites (corresponding to substitution of the OH^- groups inside the channels) during the preparation (Table 1). This is supported by the fact that such a

Table 1. Assignment of the Infrared Bands in the $1800\text{--}1150\text{ cm}^{-1}$ Range from Hydroxyapatite Pretreated Self-Supported Wafers and upon CO_2 Adsorption on Hydroxyapatite

wavenumber (cm^{-1})	assignment and location
1545–1456	bulk type B carbonate
1501–1444–1414	bulk type A carbonate
1420–1409	surface type A carbonate
1485–1385	surface $(\text{PO}_x)_s$ -carbonate
1370–1345	surface $(\text{PO}_x)_s$ -carbonate
1758–1704–1673–1664	$\nu_{\text{as}}(\text{OCO})$ of HCO^a
1610–1590	
1398 or 1390	$\nu_s(\text{OCO})$ of HCO
1180–1240	$\delta(\text{COH})$ of HCO

^aHCO = hydrogenocarbonate.

type A incorporation predominance was already reported for low carbonate contents HAp sample (<4 wt %) synthesized upon a precipitation route at high pH value.⁵⁷ The weak contributions at 1545 and 1456 cm^{-1} corresponding to B-type carbonates (Table 1).^{22,58,59} indicate that they are only present in small amounts compared to those located in the A sites. Additional contributions at $1485\text{--}1385\text{ cm}^{-1}$ and $1370\text{--}1345\text{ cm}^{-1}$ are also observed. Given that these bands are strongly influenced upon heating from RT to 623 K (see Supporting Information, Figure S2), they can be assigned to surface carbonates (Table 1). The presence of such surface carbonates results from the chemisorption of CO_2 from the atmosphere on basic O^{2-} oxide ions exposed at the surface. From the hydroxyapatite structure, the only oxygen atoms available to form carbonates belong to PO_4 groups, as those belonging to OH groups of the channels led to the formation of hydrogenocarbonates. Note that the possible close vicinity of these surface carbonates with calcium cations cannot be excluded, some oxygen atoms might bridge between phosphorus and calcium in the structure via an O atom belonging to the PO_4 group. Because those oxygen atoms are covalently bonded to phosphorus, the related surface chemisorbed carbonates will be hereafter referred to as $(\text{PO}_x)_s$ -carbonates. The fact that these surface $(\text{PO}_x)_s$ -carbonate contributions have been previously scarcely discussed in the assignment of ν_{CO} bands of hydroxyapatites is probably due to the fact that most of the spectroscopic studies dealing with this system come from the biological application field (hydroxyapatite being the main mineral component of teeth and bones)^{60–63} and that the corresponding infrared characterizations (mostly based on KBr spectra) have been discussed only in relation with bulk properties of hydroxyapatites.

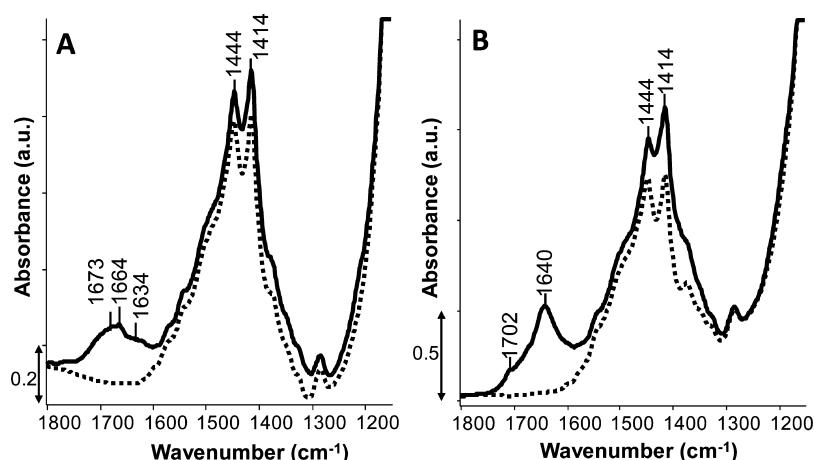


Figure 2. Influence of CO₂ adsorption in the 1800–1150 cm^{−1} area on the absolute infrared spectra of (A) nondeuterated and (B) surface deuterated HAP samples. Hydroxyapatite spectra before (dashed line) and after (solid line) exposure to CO₂ ($P_{\text{eq}}(\text{CO}_2) = 133$ Pa).

On the deuterated sample, additional $\nu_{(\text{OD})}$ contributions related to surface species are present.^{35,48} Given that the applied isotopic labeling procedure aimed at discriminating between surface and bulk OH species, bulk deuteration of the samples had to be avoided, which made surface deuteration not complete. Thus, there are still some remaining surface $\nu_{(\text{OH})}$ vibrators. Moreover, note that the two samples also differ in their final hydration state: despite a final treatment at 473 K under vacuum was achieved for the deuterated sample (section 2.2), few amounts of D₂O still remains adsorbed (contribution at 2600–2400 cm^{−1} not shown).⁴⁸

3.2. Interaction with CO₂ Followed by FTIR. **3.2.1. Adsorption Processes.** **3.2.1.1. Qualitative Description of the Modifications of the IR Spectra.** Due to the complex processes occurring upon CO₂ adsorption on HAp, the observed spectroscopic features are not simple to interpret.⁴⁸ In order to get a deeper insight into the understanding of these processes, two sets of experiments were performed in which CO₂ was contacted with a classically pretreated HAp sample (nondeuterated sample) and a surface-deuterated HAp sample. Surface deuteration is expected to improve the resolution in the $\nu_{(\text{O-H})}$, $\delta_{(\text{H}_2\text{O})}$ regions and facilitate the fine observation of the modifications occurring in the corresponding areas upon CO₂ adsorption. Moreover, such a deuteration procedure allows us to follow the influence of residual chemisorbed water on the adsorption of CO₂.

For both samples, CO₂ adsorption immediately leads to an increase or perturbation of the pre-existing carbonate and hydroxyl groups. Additional new bands are also observed in the related regions but also at 2352 cm^{−1} (not shown), this contribution being characteristic of physisorbed CO₂.

Modifications of the 1800–800 cm^{−1} Area. After introduction of an equilibrium pressure of 133 Pa of CO₂, the intensity of the pre-existing multicomponents contributions ascribed to carbonates (mainly carbonates in A sites with bands at 1444 and 1414 cm^{−1}; Figure 2A,B) seems to increase. Similar absolute spectra were already reported upon CO₂ adsorption,³⁷ from which it was concluded to the substitution of hydroxyls in A sites by carbonates in relation with the well-known affinity of hydroxyapatites toward the incorporation of structural carbonates.¹⁴ Such an easy bulk substitution process occurring at RT from solid–gas interaction is, however, surprising, and more

insights into the occurring processes can be obtained by examining the difference spectra.

Indeed, exposure of HAp samples to incremental doses of CO₂ resulted in an increase in the intensity of three multicomponent areas at about 1420, 1670, and 1210 cm^{−1} (Figure 3).

The bands at 1420 and 1409 cm^{−1}, can be assigned to the formation surface type A carbonates (Table 1) by substitution of two neighboring surface OH[−] by carbonate species. The

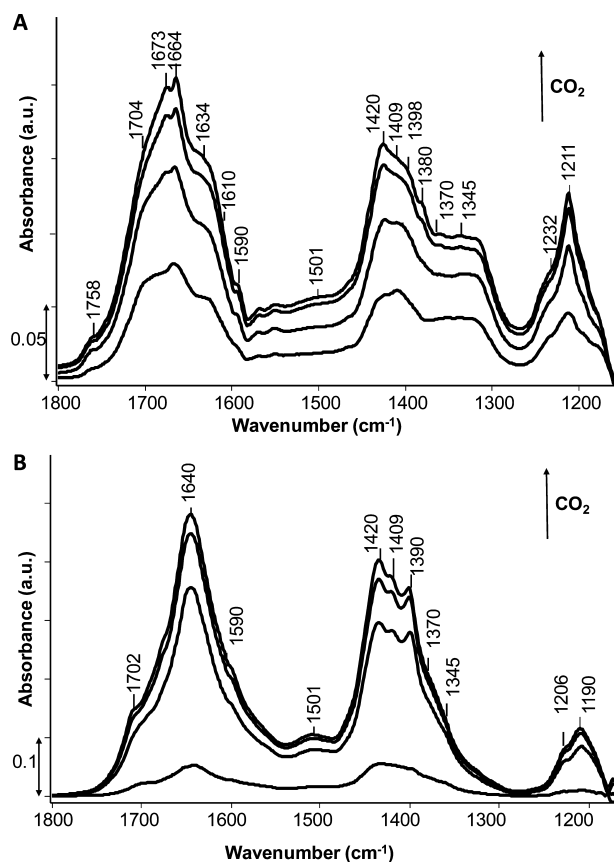


Figure 3. Difference spectra obtained after adsorption of incremental doses of CO₂ (0.5, 2.3, and 6.8 μmol and $P_{\text{eq}} = 133$ Pa) for (A) the nondeuterated and (B) the surface-deuterated HAP samples.

expected different local configurations for carbonates located in A sites at the surface compared to those already present in the bulk might explain the slight shift of these surface contributions compared to those corresponding to the original bulk type A species (1444 and 1414 cm^{-1}). This substitution process is supported by the concomitant formation of water, as illustrated by the band around 1640 cm^{-1} , ascribed to the $\delta_{(\text{H}_2\text{O})}$ bending mode of physisorbed water ($\text{CO}_2 + 2 \text{OH}^- \rightarrow \text{CO}_3^{2-} + \text{H}_2\text{O}$).

However, there is a quite important discrepancy between the intensity of this contribution at 1640 cm^{-1} and that of the corresponding $\nu_{(\text{OH})}$ broad band between 3500 and 2700 cm^{-1} that is related to physisorbed molecular water (not shown). This can be explained by the involvement of multicomponents contributions in the 1800–1600 cm^{-1} area. Focusing first on the case of the nondeuterated sample, (Figure 3A), besides the band at 1634 cm^{-1} that is precisely ascribed to $\delta_{(\text{H}_2\text{O})}$ contribution, several other contributions are observed between 1760 and 1590 cm^{-1} . They can be assigned to the $\nu_{\text{as}(\text{OCO})}$ vibration modes of hydrogenocarbonate species HCO_3^- (Table 1) observed on many basic oxides^{64–66} and resulting from the chemisorption of CO_2 on basic OH^- groups. The multiplicity of the contributions could account for different local configurations of the hydrogenocarbonate species.⁶⁶ Consistently, the associated $\nu_{\text{s}(\text{OCO})}$ contributions expected at lower wavenumber are detected, in particular at 1398 cm^{-1} (Figure 3A; Table 1). Associated $\delta_{(\text{COH})}$ and $\delta_{(\text{OCO})}$ of hydrogenocarbonate species are also observed in the 1180–1240 cm^{-1} range (Table 1) and at 833 cm^{-1} (not shown), respectively.^{67,68} Up to now, no direct evidence for hydrogenocarbonate formation on hydroxyapatite was reported, except as a transient band around 1666 cm^{-1} possibly ascribed to either “unstable CO_2 ” or to “ HCO_3 like species”, according to Cheng et al.³⁷

The adsorption of CO_2 on the deuterated surface leads to the formation of both protonated and deuterated forms not only of water ($\nu_{(\text{O-H})}$ and $\nu_{(\text{O-D})}$ broad contributions below 3500 and 2400 cm^{-1} , respectively, not shown), but also of hydrogenocarbonates. As a result, the shape and position of the maxima and relative intensities in the 1150–1800 cm^{-1} range are significantly modified compared to the adsorption of CO_2 on the nondeuterated sample (Figures 2 and 3). Due to the large red-shift ($\sim -250 \text{ cm}^{-1}$)⁶⁶ of the $\delta_{(\text{COD})}$ contribution compared to the $\delta_{(\text{COH})}$ contribution, the $\delta_{(\text{COD})}$ expected around 970 cm^{-1} is not detected. It is consistent with the fact that, despite the new involvement of $\delta_{(\text{D}_2\text{O})}$ bending mode, around 1206 cm^{-169} and of the presence of few remaining $\delta_{(\text{COH})}$ contributions, the relative intensities of the 1180–1240 cm^{-1} area are lowered on the deuterated sample compared to what is observed for the nondeuterated system. In a similar way, the shape of the multicomponents contribution in the 1770–1550 cm^{-1} area is also greatly modified. Among the $\nu_{\text{as}(\text{OCO})}$ modes, that at 1758 cm^{-1} is clearly red-shifted to 1702 cm^{-1} .³³ The red-shift of the other contributions due to deuteration together with the relative decrease in the $\delta_{(\text{H}_2\text{O})}$ contribution are responsible for the global shift of the maximum of the multicomponents contribution to 1640 cm^{-1} . Such red-shift effect also explains that the shoulder at 1398 cm^{-1} ($\nu_{\text{s}(\text{OCO})}$) now appears as a well-defined band at 1390 cm^{-1} on the deuterated sample (Figure 3A,B).

Besides bands ascribed to the formation of surface type A carbonate, water, and hydrogenocarbonate, a contribution at 1380 cm^{-1} and a broad one at 1370–1345 cm^{-1} increase on the

nondeuterated sample (Figure 3A). As reported in section 3.1, these contributions might be related to the formation of two different types of surface $(\text{PO}_x)_\text{s}$ -carbonate species (Table 1). Their formation indicates thus the involvement of basic surface O^{2-} ions belonging to phosphate groups. As reported above (Supporting Information, Figure S2 and Table 1), the associated contribution of the band at 1380 cm^{-1} is expected around 1485 cm^{-1} . Here, it could be hidden here by the overlapping of the enhanced contribution of surface type A carbonate species (intense contributions at 1420–1409 cm^{-1}). Interestingly, the contributions of the two types of surface $(\text{PO}_x)_\text{s}$ -carbonate observed on the nondeuterated sample (band at 1380 cm^{-1} and broad contribution at 1370–1345 cm^{-1}) is less visible on the deuterated sample. The broad contribution at 1370–1345 cm^{-1} is now observed only as shoulders (Figure 3B). At this stage, it is still unclear whether (i) the contributions related to these surface $(\text{PO}_x)_\text{s}$ -carbonates are partially hidden on the deuterated sample simply due to the red-shift of the $\nu_{\text{s}(\text{OCO})}$ contribution of hydrogenocarbonate upon deuteration and (ii) the decreased relative intensity of the 1380 and 1370–1345 cm^{-1} contributions could account for a decreased concentration of surface $(\text{PO}_x)_\text{s}$ -carbonate on the deuterated sample.

Modifications of the OH and OD Stretching Areas. The formation of water and hydrogenocarbonates species upon CO_2 adsorption is also expected to impact the area related to the stretching vibration modes of hydroxyl groups. Broad contributions in the 3550–3000 cm^{-1} range (or 2600–2200 cm^{-1}) associated with $\nu_{(\text{OH})}$ of H_2O (or $\nu_{(\text{OD})}$ of D_2O) are observed (not shown). Moreover, new $\nu_{(\text{OH})}$ stretching bands associated with the hydrogenocarbonate species⁶⁶ are detected (Figure 4A,B). The main $\nu_{(\text{OH})}$ stretching contribution of hydrogenocarbonate species is observed at 3616 cm^{-1} on both samples. In agreement with the expected isotopic shift ($\nu_{\text{OH}}/\nu_{\text{OD}} = 1.36$), the corresponding $\nu_{(\text{OD})}$ contribution is centered at 2665 cm^{-1} on the deuterated sample (Figure 4C). Note also that the $\nu_{(\text{OD})}$ and $\nu_{(\text{OH})}$ contributions are slightly more resolved on the deuterated sample (Figure 4B,C) than the corresponding $\nu_{(\text{OH})}$ contributions on the nondeuterated sample (Figure 4A). This may be due to the distribution of the vibrators into $\nu_{(\text{OD})}$ and $\nu_{(\text{OH})}$ contributions. To our knowledge, these modifications of the spectra in the $\nu_{(\text{OH})}$ region, although expected from the clear evidence of hydrogenocarbonate formation in the $\nu_{(\text{CO})}$ area, have not been described for the hydroxyapatite system.

More surprising is the appearance of another contribution centered at 3707 cm^{-1} . Given that a corresponding contribution was observed at 2734 cm^{-1} for the deuterated sample and the ratio 3707/2734 value (1.36) corresponds to the expected isotopic value between ν_{OH} and ν_{OD} contributions, these contributions should unambiguously be ascribed to ν_{OH} and ν_{OD} vibration modes, respectively. According to Low et al., who investigated the CaO and $\text{Ca}(\text{OH})_2$ systems, these bands could be assigned to $\nu_{(\text{Ca-O-H})}$ associated with the conversion of part of the adsorbed water into metastable brucite-like pseudolattice $\text{Ca}(\text{OH})_2$ species.⁷⁰ A close interaction of water with calcium cations has already been proposed by Bertinetti et al. from experiments aimed at studying the adsorption of water on hydroxyapatites,⁷¹ but these authors did not mention the presence of a band at 3707 cm^{-1} (region not shown in the reported spectra).

If the formation of hydrogenocarbonate species upon CO_2 adsorption unambiguously probes the presence of basic OH^-

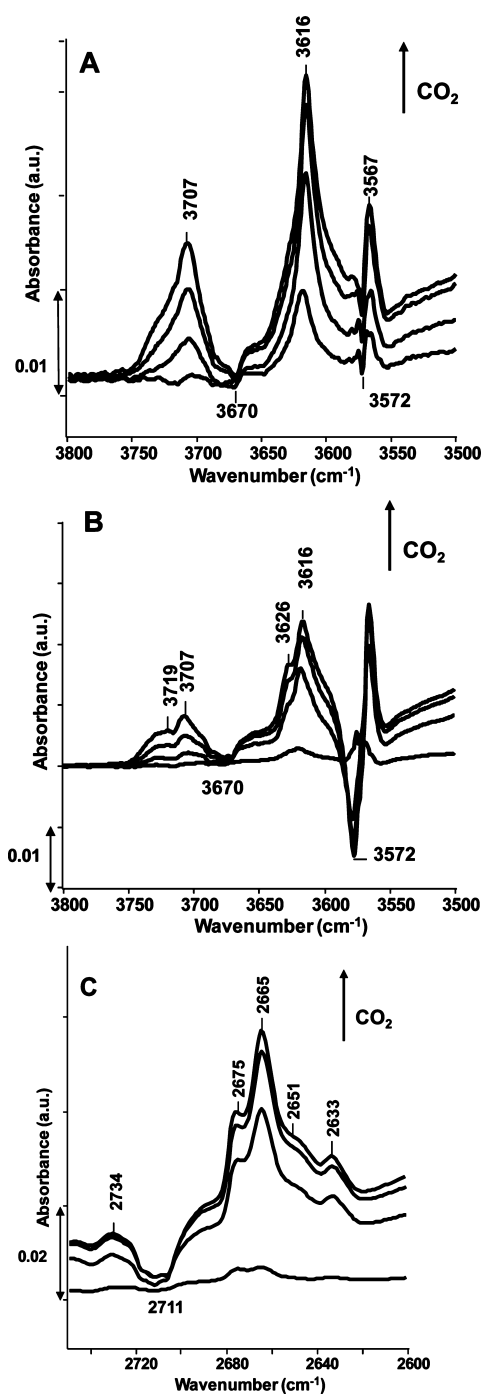


Figure 4. Difference spectra obtained after adsorption of incremental doses of CO_2 (0.5, 2.3, and 6.8 μmol and $P_{\text{eq}} = 133$ Pa) in the $\nu_{\text{(OH)}}$ areas for (A) the nondeuterated, (B) the surface-deuterated HAP samples, and (C) in the $\nu_{\text{(OD)}}$ area for the surface-deuterated HAP sample.

on the surface, due to the concomitant formation of water and brucite-like pseudolattice species, the H-bonding network is modified. This leads to the perturbation of the whole $\nu_{\text{(OH)}}$ area. Therefore, all the pre-existing $\nu_{\text{(OH)}}$ contributions associated with the basic OH^- surface that was recently proposed to be part of the 3572 cm^{-1} band,^{48,50} but also those associated with PO-H that were shown to be acidic upon CO adsorption,⁴⁸ are slightly modified (see, for instance, the negative contribution around 3670 cm^{-1} on Figure 4),

irrespectively of the involvement of the related surface species in direct interaction with CO_2 . Hence, it is not possible to formally confirm from this experiment what is the wavenumber associated with the basic OH^- groups.

3.2.1.2. Relative Intensities of the Bands on the Two Samples. If the different positions of the bands described above for the two samples are quite qualitatively consistent with the direct impact of the isotopic labeling, discrepancies between the two samples can be observed from a quantitative point of view.

The intensity of the band related to physisorbed CO_2 (2352 cm^{-1}) observed on the nondeuterated surface is about 1.5 higher than that observed on the deuterated one. This is related to a lower reactivity of the related surface toward CO_2 . Indeed, as shown from difference spectra reported on Figures 2 and 3, the bands related to the formation of surface type A carbonates and hydrogenocarbonate species are about twice more intense for the $1800\text{--}1100\text{ cm}^{-1}$ area on the deuterated sample compared with the nondeuterated sample (Figures 2 and 3). A similar tendency is observed considering the global intensities in the $\nu_{\text{(OH)}}$ and $\nu_{\text{(OD)}}$ areas ascribed to hydrogenocarbonates. In contrast, the relative amount of formed pseudolattice brucite-like species seems to decrease on the deuterated sample (band at 2734 cm^{-1} vs that at 3707 cm^{-1} ; Figure 4C vs A). Moreover, as mentioned above, the bands at 1380 and $1370\text{--}1345\text{ cm}^{-1}$ associated with surface $(\text{PO}_x)_s$ -carbonates appear also less intense on the deuterated sample compared with the nondeuterated one.

These quantitative discrepancies between the two samples might be due to their different hydration levels before the introduction of CO_2 . Indeed, as mentioned above, if for the nondeuterated sample the whole physisorbed water could be successfully removed upon the pretreatment implemented at 623 K , after the surface isotopic labeling procedure, there is still some heavy water adsorbed on the surface of the deuterated sample. It was shown that physisorbed water may decrease the accessibility to calcium cations.⁷¹ This might explain the lower tendency to form pseudolattice brucite species (band at 2734 cm^{-1}) in the deuterated sample. Such a decrease in the calcium accessibility could also result in a modification of the acid–base balance. The presence of water might also be responsible for the modification of the bands related to surface $(\text{PO}_x)_s$ -carbonates (bands at 1380 and $1370\text{--}1345\text{ cm}^{-1}$) and rather points to a relative decrease in their concentration on the hydrated sample. On the opposite water might also directly contribute to a higher surface concentration in basic OH^- that corresponds to hydroxyl groups emerging from the columns, which in turn would explain the higher ability to form both hydrogenocarbonate and surface type A carbonate. Thus, all the qualitative discrepancies between the two samples account for an enhancement of basicity of HAP surface ascribed to OH^- groups with increasing hydration level, whereas that ascribed to O^{2-} belonging to surface PO_4^{3-} groups is decreased.

3.2.1.3. Evolution of the Bands upon Exposure Time to CO_2 . Such a modification of the amount of basic sites in the presence of physisorbed water is also consistent with the evolution of the relative intensities of the infrared bands observed upon time of exposure to CO_2 for both samples (Figure 5). Indeed, as reported above, water is formed in situ upon CO_2 adsorption. This might thus explain why, as illustrated on Figure 5A,B in the case of the nondeuterated sample (similar tendency for the deuterated one), the intensity of the band at 2352 cm^{-1} ascribed to physisorbed CO_2 progressively decreases upon increasing time of exposure to

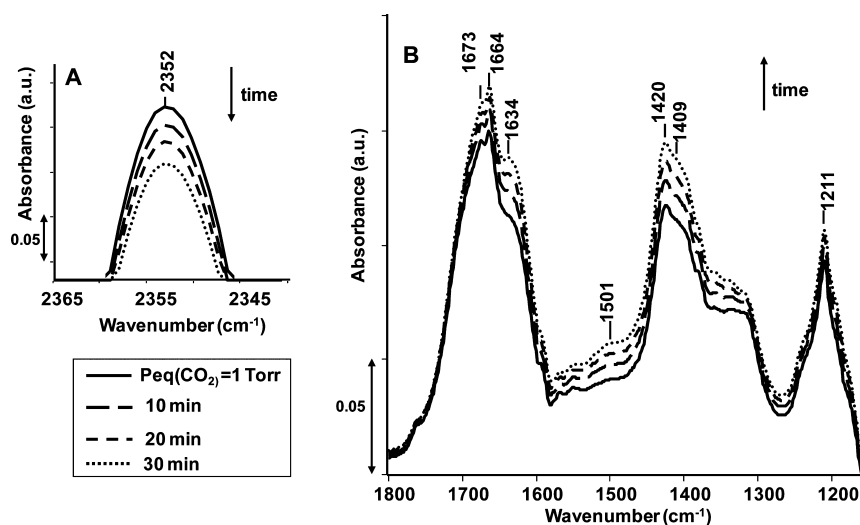


Figure 5. Influence of the time of exposure to CO_2 ($P_{\text{eq}}(\text{CO}_2) = 133 \text{ Pa}$) on the infrared spectra (A) top of the band at 2352 cm^{-1} and (B) in the $1800\text{--}1150 \text{ cm}^{-1}$ range.

CO_2 (Figure 5A), whereas the intensities of those related to hydrogenocarbonates and water (Figure 5B) simultaneously increase. Thus, physisorbed CO_2 appears to be progressively consumed at the benefit of those species. Even if such evolution of the band at 2352 cm^{-1} has previously been reported by Cheng et al.,³⁷ they proposed that the hydrogenocarbonate species might be involved only as transitory species, which is not so clear in the present study since the related bands increase together with those of surface type A carbonates. It should be noted, however, that the hydroxyapatite sample studied in this work³⁷ was under-stoichiometric ($\text{Ca}/\text{P} = 1.59$) and, thus, exhibited a lower concentration in basic OH^- groups compared to stoichiometric HAp.¹⁴ This could account for the lack of formation of hydrogenocarbonates in their study.

3.2.2. Evolution of the Spectra upon Evacuation. Upon evacuation at RT, the intensity of all the bands formed upon adsorption decreases (Figure 6). Among them, that at 3707 cm^{-1} vanishes the fastest, which is in agreement with the metastable property of the pseudolatitice brucite⁷⁰ (Figure 6A). All the bands associated with hydrogenocarbonate species ($\nu_{\text{as}}(\text{OCO})$ at 1758 , 1704 , 1673 , 1664 , 1610 , and 1590 cm^{-1} , $\nu_{\text{s}}(\text{OCO})$ at 1398 cm^{-1} , $\delta_{\text{(COH)}}$ at 1211 cm^{-1} (Figure 6B; Table 1), $\nu_{\text{(OCO)}}$ at 833 cm^{-1} (not shown), and $\nu_{\text{(OH)}}$ at 3616 cm^{-1} (Figure 6A)) and to surface $(\text{PO}_x)_s$ -carbonate $\nu_{\text{(CO)}}$ ($1370\text{--}1345 \text{ cm}^{-1}$; Table 1) also decrease rapidly. On the other hand, the band associated with water (see the relative evolution of the $\delta_{\text{(H}_2\text{O)}}$ band at 1634 cm^{-1} compared to the $\nu_{\text{as}}(\text{OCO})$ contribution related to hydrogenocarbonates) and that at 1409 cm^{-1} seem to be more persistent (Figure 6B). Surprisingly, the two bands ascribed to surface type A carbonates (at 1420 and 1409 cm^{-1} ; Table 1) do not seem to decrease simultaneously. As shown from a comparison with the inset (b) in Figure 1, this could be due to the overlapping with a band corresponding to a contribution of surface $(\text{PO}_x)_s$ -carbonates that is expected at 1485 cm^{-1} (Table 1). Thus, the more rapid desorption of surface $(\text{PO}_x)_s$ -carbonates compared to surface type A carbonates could finally explain the apparent fastest decrease of the 1420 cm^{-1} contribution compared to that of the band at 1409 cm^{-1} . Water and surface type A carbonate that are both formed upon substitution process ($\text{CO}_2 + 2\text{OH}^- \rightarrow \text{CO}_3^{2-} + \text{H}_2\text{O}$) are the most stable species. The complete recovery of the

initial spectrum is finally achieved after outgassing at 523 K (not shown). Similar trends could also be observed on the deuterated sample.

3.2.3. Synopsis and Evaluation of CO_2 as a Tool to Characterize the Basicity of Hydroxyapatites. Hydroxyapatites are bulk materials that naturally exhibit bulk carbonates as well as surface carbonates and hydrogenocarbonates. This leads to a quite complex infrared spectrum. The assignments of the related bands present in the $1800\text{--}1150 \text{ cm}^{-1}$ region that are gathered in Table 1 are based on the changes of the related infrared bands observed (i) upon thermal treatment at 623 K that impacts the relative concentration of the surface chemisorbed species, (ii) upon thermal treatment at 873 K that also impacts the bulk species (Supporting Information, Figure S3), and (iii) upon the adsorption–desorption processes of CO_2 .

This CO_2 adsorption–desorption process monitored with infrared spectroscopy reveals the presence of basic OH^- groups and O^{2-} of surface PO_4^{3-} groups leading to the formation of hydrogenocarbonates and surface $(\text{PO}_x)_s$ -carbonates, respectively. Thus, CO_2 enables to qualitatively probe the nature of basic sites, even though no direct identification of the $\nu_{\text{(OH)}}$ spectroscopic signature of basic OH^- can be achieved due to the complexity of the adsorption process. Moreover, it is evidenced that water formed from substitution of surface OH^- groups by carbonates (surface type A carbonates) or even pre-existing on the surface before introducing CO_2 increases the basicity ascribed to hydroxyl groups (increase of bands related to hydrogenocarbonates), whereas it decreases that ascribed to O^{2-} species (decrease of the bands related to surface $(\text{PO}_x)_s$ -carbonates). This result is of peculiar importance since it might impact the reactivity of the hydroxyapatites under reaction conditions, depending on the pretreatment conditions and on the amount of water formed as a product or added in the reaction feed.

However, if the formation of these surface $(\text{PO}_x)_s$ -carbonates and hydrogenocarbonates directly probes the Lewis basicity of HAp surfaces toward CO_2 , the formation of surface type A carbonate and associated water is not simply related to surface basicity and rather accounts for the restructuring of the surface. Such substitution process is facilitated by the well-known structural affinity of hydroxyapatites for carbonation,

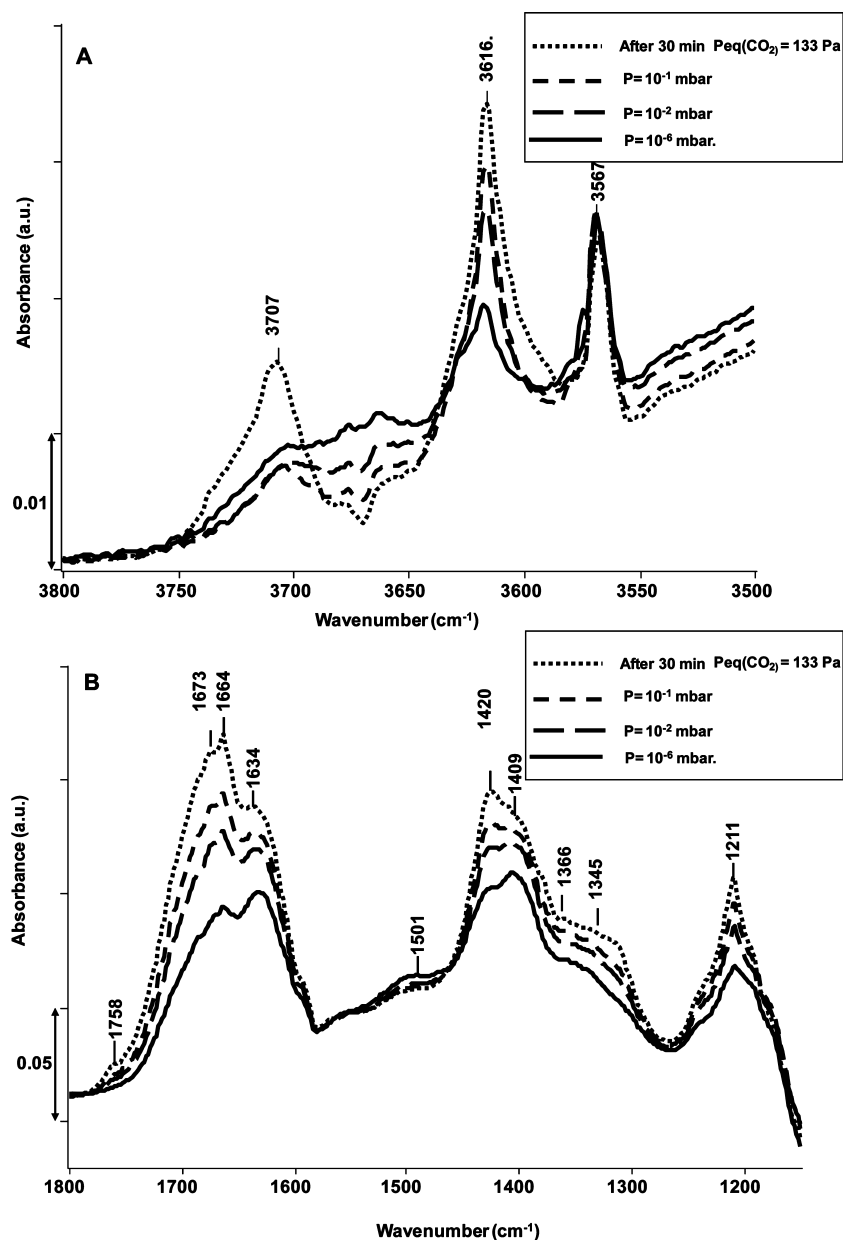


Figure 6. Difference spectra obtained after introduction of CO_2 ($P_{\text{eq}}(\text{CO}_2) = 133 \text{ Pa}$) and subsequent outgassing at RT (A) in the $\nu_{(\text{OH})}$ area and (B) in the $1800\text{--}1150 \text{ cm}^{-1}$ range.

but at molecular level, it requires a convenient proximity between two OH^- groups. DFT calculations are scheduled to investigate what are the structural surface properties leading either to the formation of stable hydrogenocarbonate species or to the substitution process, resulting in the formation of surface type A carbonates.

From the desorption process, among the species formed upon CO_2 adsorption, those probing the basic sites are eliminated at quite low temperature. This underlines that the hydroxyapatite system exhibits weak basic sites. However, it is not possible to go further using CO_2 as a quantitative tool to evaluate the density and strength of basic sites, respectively, as classically deduced following the thermal desorption of CO_2 . Indeed, despite the relative lower stability of hydrogenocarbonates and surface $(\text{PO}_x)_s$ -carbonates compared to water and surface type A carbonates, it is not possible to experimentally discriminate the CO_2 desorbing from OH^- and O^{2-} basic sites

from that formed from the reversible substitution on surface type A sites ($\text{CO}_3^{2-} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 2\text{OH}^-$).

Moreover, considering that the recovery of the initial surface is obtained upon outgassing at quite limited temperature (523 K), it is even questionable if all the data based on thermodesorption techniques, that are commonly implemented at least up to 873 K, account for the production of CO_2 associated with the decomposition of bulk type A and B carbonates⁷² rather than for the lone surface basicity characterization. Indeed, from the present results and additional data reported in Supporting Information, Figure S3, upon thermal treatment, the relative amount of bulk carbonates located in A sites decreases first (as mentioned above) before increasing up to 873 K, whereas most type B carbonates decompose below 873 K. These data are consistent with the studies of Holcomb et al.⁵⁹ in which it was concluded that the thermal

decomposition of bulk type B carbonates proceeds via their transient moving to A sites.

3.3. Interaction with Acetylene Followed by DRIFTS.

Whereas in the case of CO_2 adsorption, the involvement of basic PO_4^{3-} and OH^- could be probed by the formation of carbonate and hydrogenocarbonate, respectively, the possible involvement of the former groups is more difficult to investigate in the case of C_2H_2 due to the saturation of the $\nu_{(\text{P}-\text{O})}$ bands. From preliminary data obtained by FTIR set up (not reported), and as observed for CO_2 adsorption, the associated overtone and combinations bands are not either effective: their lack of sensitivity is reinforced by the low proportions of surface phosphates groups. Moreover, these contributions ($1900\text{--}2200\text{ cm}^{-1}$) overlap with $\nu_{\text{C}\equiv\text{C}}$ contribution of acetylene (expected in the $1900\text{--}2000\text{ cm}^{-1}$ range). To overcome these difficulties, DRIFTS was preferentially used: two series of DRIFTS experiments were performed, on pure HAP sample to examine the adsorption mode of acetylene (same features observed by FTIR, not reported here) and on diluted HAP sample to investigate the involvement of PO_4^{3-} species.

3.3.1. Adsorption Process. **3.3.1.1. HAP Sample.** The presence of acetylene in the flow rapidly results in the formation of new IR bands as seen on the difference spectra (subtraction of the spectrum of acetylene in gas phase and of the spectrum recorded just before introduction of acetylene in the flow) reported in Figure 7.

An asymmetrical shaped band is observed at 1949 cm^{-1} (Figure 7A). It is related to the $\nu_{(\text{C}\equiv\text{C})}$ of adsorbed acetylene ($\nu_{(\text{C}\equiv\text{C})}$ contribution in gas phase at 1974 cm^{-1}).³⁸ Moreover, in the $3300\text{--}3000\text{ cm}^{-1}$ range ascribed to the $\nu_{(\text{C}-\text{H})}$ stretching mode of adsorbed acetylene (Figure 7B), a maximum is detected at 3237 cm^{-1} , whereas a contribution of low intensity and a broad shoulder are observed at 3281 cm^{-1} and around 3150 cm^{-1} , respectively ($\nu_{3(\text{C}-\text{H})} = 3287\text{ cm}^{-1}$ in the gas phase³⁸). The $\nu_{(\text{C}-\text{H})}$ contributions could be decomposed evidencing five components at 3281 , 3237 , 3214 , 3178 , and 3099 cm^{-1} (Figure 7C). In the absence of any strong basic sites on the HAP system, only asymmetric $\nu_{(\text{C}-\text{H})}$ contributions could be observed by infrared spectroscopy.^{73,74} These five asymmetric $\nu_{\text{C}-\text{H}}$ contributions rapidly reach their maximum intensity and no significant relative evolution is observed during the adsorption process after 6 min on stream (Figure 8).

Starting from the spectrum recorded just before acetylene adsorption (shown in Supporting Information, Figure S1), the adsorption of acetylene also leads to the modification of the spectra in the $\nu_{(\text{OH})}$ region (Figure 9). From the absence of appearance of new bands, it can be concluded that the surface is not able to deprotonate the acetylene molecule. However, as seen on the difference spectra, the pre-existing $\nu_{(\text{OH})}$ bands are perturbed. As reminded above, besides the band at 3572 cm^{-1} , the infrared spectrum of hydroxyapatite exhibits several PO-H contributions in this region (inset (a) in Figure 1). Among these PO-H vibrators, only that at 3657 cm^{-1} is clearly not modified upon acetylene adsorption (Figure 9). This supports its previously proposed assignment to bulk HPO_4^{2-} species that was based on isotopic labeling of HAP surface experiments.⁴⁸ The other PO-H contributions at 3714 , 3680 , and 3670 cm^{-1} are clearly perturbed. It is noteworthy that these PO-H bands are perturbed to a various extent: there is no relation between the relative intensity of the PO-H contributions on the spectrum recorded just after pretreatment (similar relative intensities than reported in inset a of Figure 1) and that of the corresponding negative bands on difference

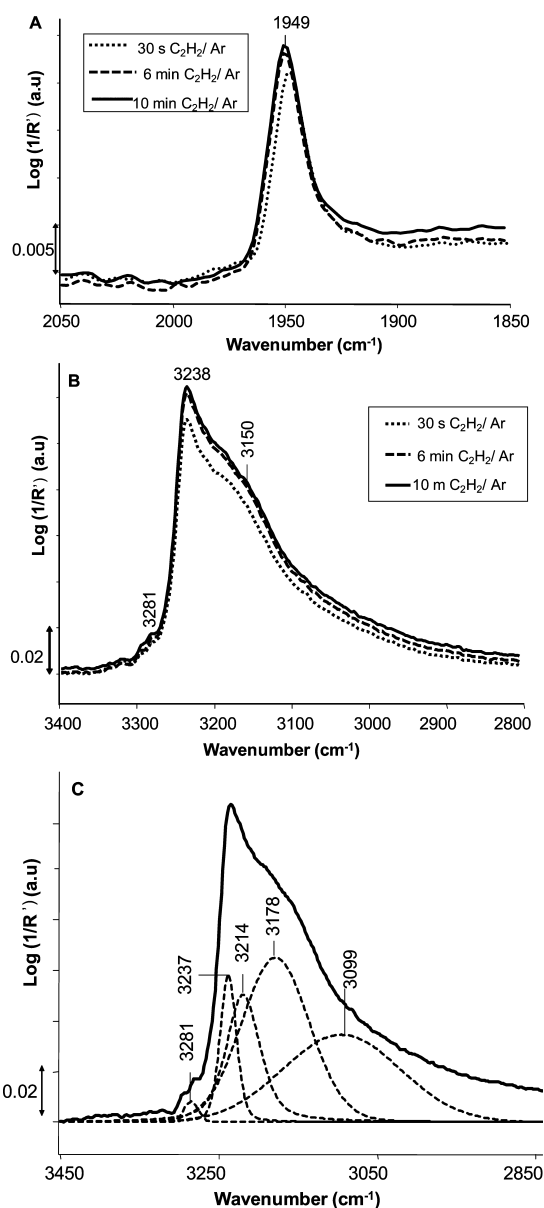


Figure 7. DRIFT difference spectra recorded after increasing exposure time to $\text{C}_2\text{H}_2/\text{Ar}$ flow at RT in the range of the (A) $\nu_{(\text{C}\equiv\text{C})}$ contributions, (B) $\nu_{(\text{C}-\text{H})}$ contributions, and (C) decomposition of the $\nu_{(\text{C}-\text{H})}$ region exemplified on the spectrum recorded after 6 min of contact time with $\text{C}_2\text{H}_2/\text{Ar}$ flow.

spectra obtained after C_2H_2 adsorption (Figure 9). In particular, the contribution at 3680 cm^{-1} is more impacted than that at 3670 cm^{-1} . This could be due to the involvement of these surface PO-H contributions in different adsorption modes of acetylene and to different interaction strengths with the corresponding adsorbates. From previous results obtained from CO adsorption at low temperature,⁴⁸ all these bands were associated with acidic surface POH species. In addition, the band at 3572 cm^{-1} is also clearly perturbed (Figure 9). This definitely points toward the involvement of a few surface O-H vibrators in this band that is dominated by bulk O-H vibrators, as suggested in a previous paper.⁴⁸ Thus, both acidic PO-H and basic OH^- groups are involved in the interaction of acetylene with the HAP surface.

3.3.1.2. Diamond Diluted HAP Sample. The inert nature of diamond allowed us to perform the same thermal pretreatment

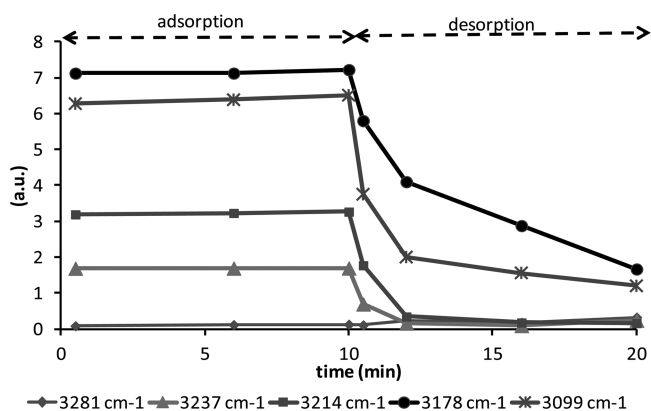


Figure 8. Evolution of the areas of the $\nu_{(\text{C-H})}$ contributions during the adsorption and desorption processes under $\text{C}_2\text{H}_2/\text{Ar}$ and Ar flow, respectively. The curve-fit of the spectra was performed with the Levenberg–Marquardt method from the Galactic Grams software.

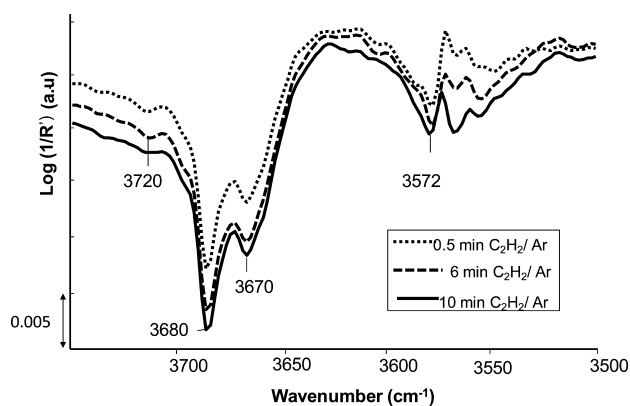


Figure 9. Difference DRIFT spectra in the $\nu_{(\text{OH})}$ area recorded after increasing time under the $\text{C}_2\text{H}_2/\text{Ar}$ flow.

before introducing C_2H_2 on the diamond diluted HAp sample. Under these conditions, despite the dilution, DRIFT spectroscopy still appears sensitive enough to detect the adsorption of acetylene on the HAp surface via the observation of $\nu_{(\text{C-H})}$ bands and the perturbation of the HAp surface in the $\nu_{(\text{O-H})}$ area (see Supporting Information, Figure S4). From the similar shape of the spectrum obtained for pure HAp sample in the $\nu_{(\text{C-H})}$ region (Figure 7B), similar interaction modes are involved in both series of experiments. However, in the 1500–900 cm^{-1} range, there is no clear evidence of any perturbation of vibrator of surface PO_4^{3-} species (not shown). This can be attributed either to a limited PO_4^{3-} surface density and to the limited basicity of the O^{2-} of these PO_4^{3-} groups. Concerning this first explanation, from the morphology of the related sample (elongated rod, see Supporting Information, Figure S5), (100) planes are preferentially exposed. These planes expose either Ca or phosphate enriched terminations.⁷⁵ In the present case, as reported elsewhere for the same sample, the surface exposes only a few Ca^{2+} cations accessible to the adsorption of basic molecule (CO).⁴⁸ This rather points to phosphate enriched terminations. Even though it is well established that the surface terminated phosphate groups are protonated to balance the surface charge^{53,54} (as illustrated by the detection of several surface acidic PO-H vibrators^{23,48,76}), the results obtained from CO_2 adsorption also revealed the presence of basic O^{2-} sites of weak basicity. It can thus be inferred that these surface O^{2-} of PO_4^{3-} groups sites are not basic enough to

be involved in the adsorption process of the acetylene molecule ($\text{p}K_a$ of C_2H_2 (~ 25) higher than that of CO_2 (~ 6.4)), whereas surface OH^- are. The weaker basic strength of O^{2-} of PO_4^{3-} groups compared to that of the OH^- groups might also account for the absence of any basic reactivity of the octacalcium phosphate compound $\text{Ca}_8(\text{HPO}_4)_2(\text{PO}_4)_4 \cdot 5\text{H}_2\text{O}$ ⁷⁷ that is deprived of any basic OH groups.²³

3.3.2. Desorption Process. One can observe a slight shift of the maximum of the $\nu_{(\text{C}\equiv\text{C})}$ contributions from 1949 to 1944 cm^{-1} upon progressive desorption (Figure 10A). This is indicative of different interaction strengths of the various adsorption modes involved.

More information can be obtained from the $\nu_{(\text{C-H})}$ region. Compared to the difference spectrum recorded after 10 min under C_2H_2 flow (end of the adsorption process), the elimination of C_2H_2 in the Ar flow rapidly results in a decrease

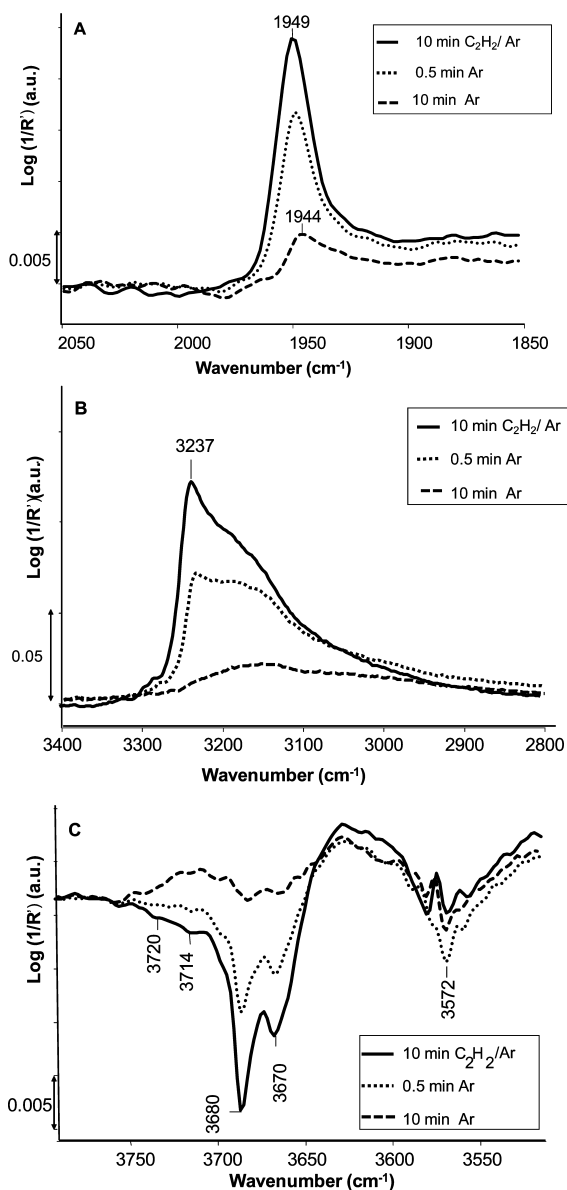


Figure 10. Desorption process under an Ar flow followed by difference DRIFT spectra (reference spectrum recorded just before the introduction of C_2H_2 in the flow) in the (A) $\nu_{(\text{C}\equiv\text{C})}$, (B) $\nu_{(\text{C-H})}$, and (C) $\nu_{(\text{OH})}$ regions.

of all the $\nu_{(\text{C-H})}$ components associated with adsorbed C_2H_2 species (Figure 8). Most of the species are removed after 10 min under an Ar flow. However, as depicted by the modification of the shape of the $\nu_{(\text{C-H})}$ components envelop (Figure 10B), the different components are not impacted in a similar way upon desorption. Indeed, from the decomposition of the spectra, the intensity of the contributions at 3237, 3214, and 3099 cm^{-1} decreases by about 50% within 0.5 min, whereas the intensity of the bands at 3281 and 3178 cm^{-1} decreases by about 20% only (Figure 8).

The acetylene desorption from the HAp surface can also be investigated by following the progressive release of the POH and OH^- groups (Figure 10C). The surface PO-H vibrators are progressively released and remain very slightly perturbed after 10 min under Ar flow. By contrast, the perturbation of the O-H vibrators (3572 cm^{-1}) is more persistent, with even a transient enhanced perturbation after 0.5 min on stream.

3.3.3. Adsorption Modes of C_2H_2 on HAp Surface. All these features are indicative of the involvement of various adsorption modes of acetylene on the HAp surface and of the presence of sites of different strengths.

Many adsorption modes for acetylene interacting with oxide surfaces have been described in the literature.^{38,78} In particular, several nondissociative adsorption modes have been reported: the molecule can adsorb either parallel to the surface via the interaction of the $\text{C}\equiv\text{C}$ acetylenic bond (referred to as π complex interaction), or perpendicular to the surface via the interaction of one hydrogen (referred to as σ complex interaction).⁷⁸ For these two adsorption modes, acetylene adsorbs on an acid and a basic site, respectively. A third nondissociative adsorption mode involving the acid and basic sites of an acid–base pair exists as well. This nondissociative adsorption mode of C–H acidic molecules such as acetylene, propyne, or *tert*-butylacetylene on an acid–base pair has been already reported to occur either on Lewis acid sites for MgO or on weak Brønsted acid sites of silica.³⁹ Finally, a dissociative adsorption mode can also be considered in the presence of strong basic sites. The dissociation of the C–H acidic molecule on such a strong basic site may result in the formation of an OH^- group, the acetylide group being in interaction with the acid site.⁴⁰

From the absence of perturbation of P–O vibrators upon acetylene adsorption, the basic sites likely to be involved in acetylene interaction are basic OH^- groups. We also evidenced that acidic PO–H are also perturbed, and they must thus be involved as acid sites. The additional involvement of calcium cations is not straightforward since we already reported that there are only a few Ca^{2+} cations accessible to the adsorption of a basic molecule (CO),⁴⁸ which is in agreement with the absence of Lewis acid sites probed by pyridine adsorption.⁵²

As reported above, no deprotonation process of acetylene occurs on HAp. Thus, based on the observation of five different $\nu_{(\text{C-H})}$ contributions, the participation of the three nondissociative adsorption modes of acetylene on hydroxyapatites is very likely (Figure 11). Contrary to the π interaction for which the two C–H vibrators are equivalent, in the σ or in acid–base pair interactions, the two C–H bonds are not similarly perturbed, leading to two distinct C–H vibrators. Such a coupling of two series of vibrators also explains that the stability of the different $\nu_{(\text{C-H})}$ adsorbates upon the desorption process does not strictly fit the order that could be expected considering that at first order, the higher the red-shift of the C–H vibrators of the adsorbates compared to the value in the gas

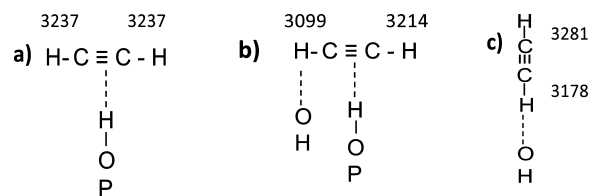


Figure 11. nondissociative adsorption modes of acetylene on hydroxyapatites and corresponding $\nu_{(\text{C-H})}$ values.

phase, the stronger is the involved interaction. Based on these data and on the evolution of the PO–H and O–H surface vibrators upon the desorption process, the following assignments are proposed:

- (i) The π complex interaction leads to only one $\nu_{(\text{C-H})}$ vibrator of the acetylene molecule: none of the two protons being directly involved in the interaction (Figure 11a). Hence, the $\nu_{(\text{C-H})}$ vibrator is only slightly perturbed compared to that in the gas phase ($\nu_{(\text{C-H})} = 3287 \text{ cm}^{-1}$). In this case, a low red-shift is expected and the band at 3237 cm^{-1} could account for such a π complex adsorption mode.
- (ii) A nondissociative interaction of C_2H_2 with an acid–base pair (Figure 11b) leads to two distinct $\nu_{(\text{C-H})}$ bands. The contribution of the free C–H vibrator must be shifted to a value close to that found in the π complex interaction, whereas the second contribution must be shifted to lower wavenumbers due to being more perturbed. Taking into account the quite rapid recovery of PO–H vibrators compared to O–H ones (Figure 10C), this adsorption mode appears to be less stable than the σ type interaction. The two components at 3214 and 3099 cm^{-1} were found to be affected similarly during the desorption step (Figure 8) and more rapidly recovered than the other coupled contributions at 3281 and 3178 cm^{-1} . Thus, the contributions at 3214 and 3099 cm^{-1} are ascribed to the $\nu_{(\text{C-H})}$ vibrators of C_2H_2 interacting with an acid–base pair (POH–OH) of HAp.
- (iii) The σ complex interaction (Figure 11c) leads to two different $\nu_{(\text{C-H})}$ vibrators. The vibrator farthest from the surface should correspond to the less red-shifted vibration, whereas the second vibrator is expected to be more perturbed. Moreover, considering the more rapid release of the adsorbates in an adsorption mode involving POH species compared to that involving only OH^- groups, the two coupled $\nu_{(\text{C-H})}$ vibrators are also expected to be the more stable upon desorption process. The contributions at 3281 and 3178 cm^{-1} are thus assigned to the $\nu_{(\text{C-H})}$ vibrators of C_2H_2 adsorbed on the HAp surface via the σ complex interaction.

4. CONCLUSION

Hydroxyapatite is a catalytic system that is used for basic or acid–base reactions. In the present work, the nature of the basic sites possibly involved in catalytic reactions^{8,15,18–22} is discussed by following the interaction of the pretreated surface with acidic molecules, CO_2 , and acetylene likely to reveal their Lewis and Brønsted basic sites, respectively. After pretreatment at 623 K, the surface is cleaned up from surface adsorbates (water, hydrogenocarbonate, and carbonate) and only few surface carbonates are still present.

Upon adsorption of CO₂, both hydrogenocarbonate and surface (PO_x)_s-carbonate species are formed, thus probing the presence of OH[−] and O^{2−} basic sites, respectively. The additional formation of surface type A carbonates accounts for the peculiar ability of the hydroxyapatite framework for substitution process. The water formed upon this substitution process as well as the presence of pre-existing physisorbed water associated with the isotopic labeling of the surface by D₂O promotes the substitution ability, but also modifies the basic properties, favoring the formation of hydrogenocarbonates at the expense of surface (PO_x)_s-carbonates. The modifications of the HAp surface resulting from CO₂ adsorption are reversible after thermal treatment at only 523 K, accounting for the weak basicity of the system. Based on this result, CO₂ cannot be considered as a reliable quantitative probe to evaluate the basicity upon thermodesorption experiments that are usually implemented up to 873 K since, in these conditions, CO₂ release accounts for the thermal decomposition of bulk carbonates rather than for the lone desorption from surface basic sites.

Acetylene is more convenient to evaluate the Brønsted basicity that is involved in real catalytic reactions. The perturbation of the 3572 cm^{−1} band upon acetylene adsorption is clearly a proof that this band, associated with bulk OH[−] from the channels, also includes the surface OH[−]. These basic OH[−] groups are involved in two nondissociative adsorption modes of acetylene. As the PO₄^{3−} groups are not perturbed by C₂H₂, it is thus concluded that basic OH[−] groups are the most likely basic active sites involved in catalytic reactions. Associated with acidic POH groups, these OH sites might be responsible for the atypical properties of hydroxyapatites observed in many catalytic reactions, as will be illustrated in a forthcoming paper dealing with operando characterizations of HAPs.

■ ASSOCIATED CONTENT

● Supporting Information

(S1) DRIFT spectrum of the Hap sample after pretreatment at 623 K under He flow. (S2) Comparison of hydroxyapatite FTIR spectrum recorded from self-supported pellet after pretreatment at 523 K and of KBr spectrum. (S3) DRIFT spectra after thermal treatment at different temperature to follow the relative stability of type A and type B carbonates. (S4) Difference DRIFT spectra recorded after 10 min of introduction of C₂H₂ in Ar flow on hydroxyapatite diluted in diamond. (S5) TEM picture of hydroxyapatite. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We would like to acknowledge the support of Sangi Co. Ltd., Tokyo.

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